

Exploratory Data Analysis on the Automobile Dataset

This notebook performs data cleaning and exploratory data analysis (EDA) on an automobile dataset. The goal is to clean, sanitise, and analyse the data in order to uncover patterns and relationships between vehicle features such as price, fuel efficiency, and engine characteristics.

The analysis follows a structured approach including data cleaning, handling missing values, and exploring key relationships through visualisations.

```
In [39]: # Import Libraries

import numpy as np
import pandas as pd
import seaborn as sns

from datetime import datetime
import matplotlib.pyplot as plt
%matplotlib inline
```

```
In [40]: # Load the automobile dataset
automobiles_df = pd.read_csv('automobile.txt', na_values='?')

automobiles_df.head()
```

Out[40]:

	symboling	normalized-losses	make	fuel-type	aspiration	num-of-doors	body-style	drive-wheels	engine-location
0	3	NaN	alfa-romero	gas	std	two	convertible	rwd	front
1	3	NaN	alfa-romero	gas	std	two	convertible	rwd	front
2	1	NaN	alfa-romero	gas	std	two	hatchback	rwd	front
3	2	164.0	audi	gas	std	four	sedan	fwd	front
4	2	164.0	audi	gas	std	four	sedan	4wd	front

5 rows × 26 columns



Data Cleaning

The dataset contains missing values represented as '?' which were converted to NaN during loading.

The following steps were performed:

- Removed unnecessary columns
- Removed duplicate rows
- Discarded rows containing missing values
- Converted relevant columns to appropriate numeric data types

These steps ensure the dataset is clean, consistent, and suitable for analysis.

```
In [41]: automobiles_df = automobiles_df.drop(['normalized-losses', 'symboling'], axis=1)
```

Remove any duplicate rows

```
In [42]: automobiles_df = automobiles_df.drop_duplicates()
```

Remove rows with missing data

Some automobiles in the database have missing values which implies that their values have not been recorded or some information is missing. Discard such entries from the dataframe.

```
In [43]: automobiles_df = automobiles_df.dropna()
```

Change columns with numerical data column to an integer data type using numpy's `int64` method.

```
In [44]: numeric_cols = [  
    'wheel-base', 'length', 'width', 'height', 'curb-weight',  
    'engine-size', 'bore', 'stroke', 'compression-ratio',  
    'horsepower', 'peak-rpm', 'city-mpg', 'highway-mpg', 'price'  
]  
  
automobiles_df[numeric_cols] = automobiles_df[numeric_cols].astype('int64')
```

Missing Data

Missing values were identified in the dataset and represented as '?'. These were converted to NaN and removed entirely as per the task instructions.

Removing rows with missing values ensures the integrity and consistency of the dataset, although it may reduce the overall dataset size.

Finding Certain Categories

Locate all automobiles in the "hatchback" genre.

```
In [45]: hatchback_cars = automobiles_df[automobiles_df['body-style'] == 'hatchback']  
hatchback_cars.head()
```

```
Out[45]:
```

	make	fuel-type	aspiration	num-of-doors	body-style	drive-wheels	engine-location	wheel-base	length	width
2	alfa-romero	gas	std	two	hatchback	rwd	front	94	171	65
18	chevrolet	gas	std	two	hatchback	fwd	front	88	141	60
19	chevrolet	gas	std	two	hatchback	fwd	front	94	155	63
21	dodge	gas	std	two	hatchback	fwd	front	93	157	63
22	dodge	gas	std	two	hatchback	fwd	front	93	157	63

5 rows × 24 columns



Insight: Hatchback vehicles form a distinct subset within the dataset, allowing for focused analysis on this category.

Now onto the exploration

Identify relationships between variables (features)

The main goal here is to identify and create relationships that can help you formulate ideas. We have defined questions to help you identify some relationships to explore.

Which are the 5 most expensive cars?

How do the most expensive and cheapest cars compare? Exploring the most expensive cars highlights if some automobiles are worth the money spent on them based on their fuel economy (mpg or miles per gallon)

```
In [46]: top_5_expensive = automobiles_df.sort_values(by='price', ascending=False).head(5)  
top_5_expensive
```

Out[46]:

	make	fuel-type	aspiration	num-of-doors	body-style	drive-wheels	engine-location	wheel-base	length	wic
74	mercedes-benz	gas	std	two	hardtop	rwd	front	112	199	
16	bmw	gas	std	two	sedan	rwd	front	103	193	
73	mercedes-benz	gas	std	four	sedan	rwd	front	120	208	
128	porsche	gas	std	two	convertible	rwd	rear	89	168	
17	bmw	gas	std	four	sedan	rwd	front	110	197	

5 rows × 24 columns



Insight: The most expensive cars tend to have larger engines and higher performance specifications. However, they do not necessarily offer better fuel efficiency, suggesting a trade-off between cost and economy.

Which manufacturer builds the most fuel efficient vehicles?

Compare the average mpg for each vehicle manufacture's vehicles and create a bar plot

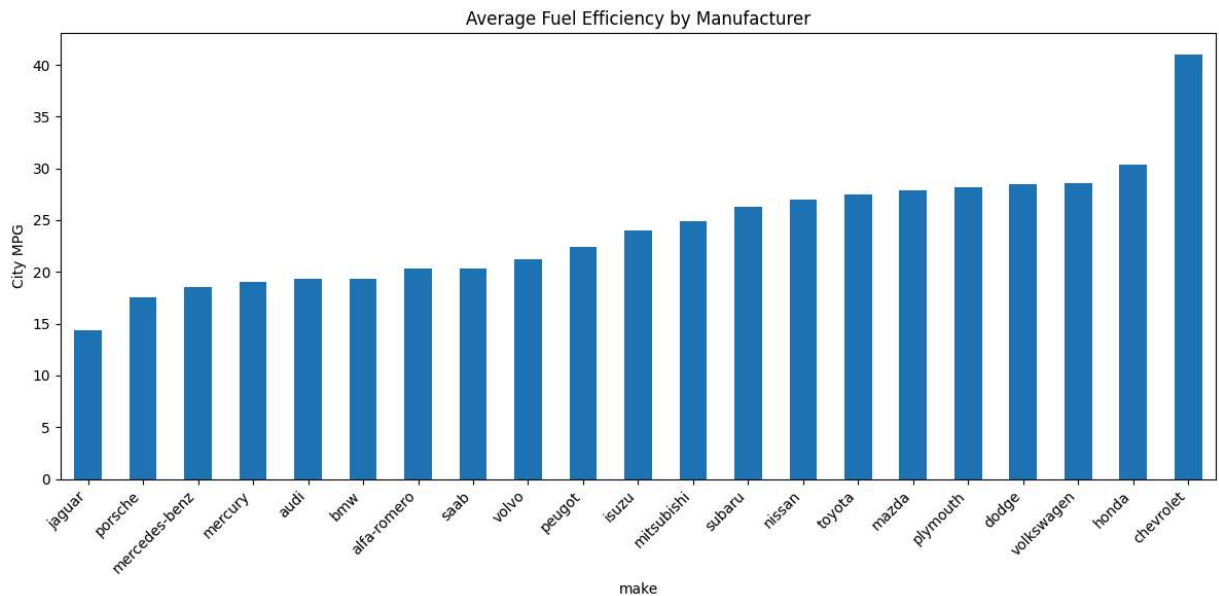
```
In [47]: avg_mpg = automobiles_df.groupby('make')['city-mpg'].mean().sort_values()

plt.figure(figsize=(12,6))

avg_mpg.plot(kind='bar')
plt.title('Average Fuel Efficiency by Manufacturer')
plt.ylabel('City MPG')

plt.xticks(rotation=45, ha='right')
plt.tight_layout()

plt.show()
```



Insight: Certain manufacturers consistently produce more fuel-efficient vehicles, highlighting differences in engineering focus and design priorities across brands.

Which vehicles have the largest engine capacity.

Sort the dataframe based on the engine-size column.

```
In [48]: largest_engines = automobiles_df.sort_values(by='engine-size', ascending=False)
largest_engines.head()
```

Out[48]:

	make	fuel-type	aspiration	num-of-doors	body-style	drive-wheels	engine-location	wheel-base	length	width
49	jaguar	gas	std	two	sedan	rwd	front	102	191	70
73	mercedes-benz	gas	std	four	sedan	rwd	front	120	208	71
74	mercedes-benz	gas	std	two	hardtop	rwd	front	112	199	72
47	jaguar	gas	std	four	sedan	rwd	front	113	199	69
48	jaguar	gas	std	four	sedan	rwd	front	113	199	69

5 rows × 24 columns

Insight: Vehicles with larger engine sizes are typically associated with higher performance and cost, but may also result in lower fuel efficiency, which suggests a strong engineering trade-off between performance and fuel efficiency.

Which vehicle manufacturer has the most car models in the dataset

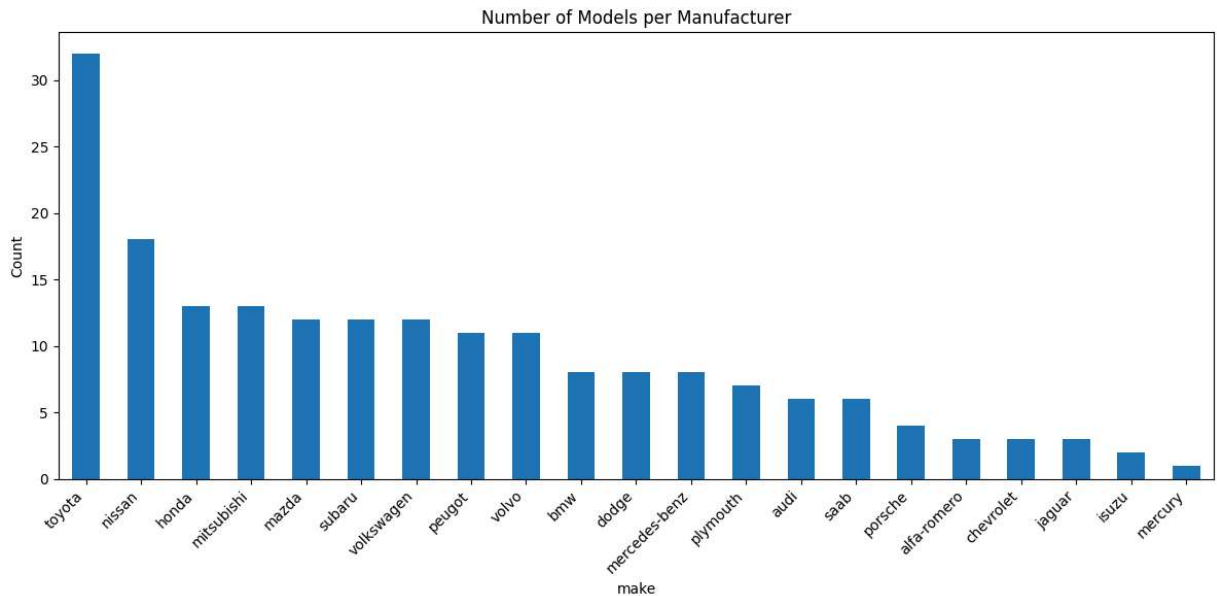
```
In [49]: manufacturer_counts = automobiles_df['make'].value_counts()

plt.figure(figsize=(12,6))

manufacturer_counts.plot(kind='bar')
plt.title('Number of Models per Manufacturer')
plt.ylabel('Count')

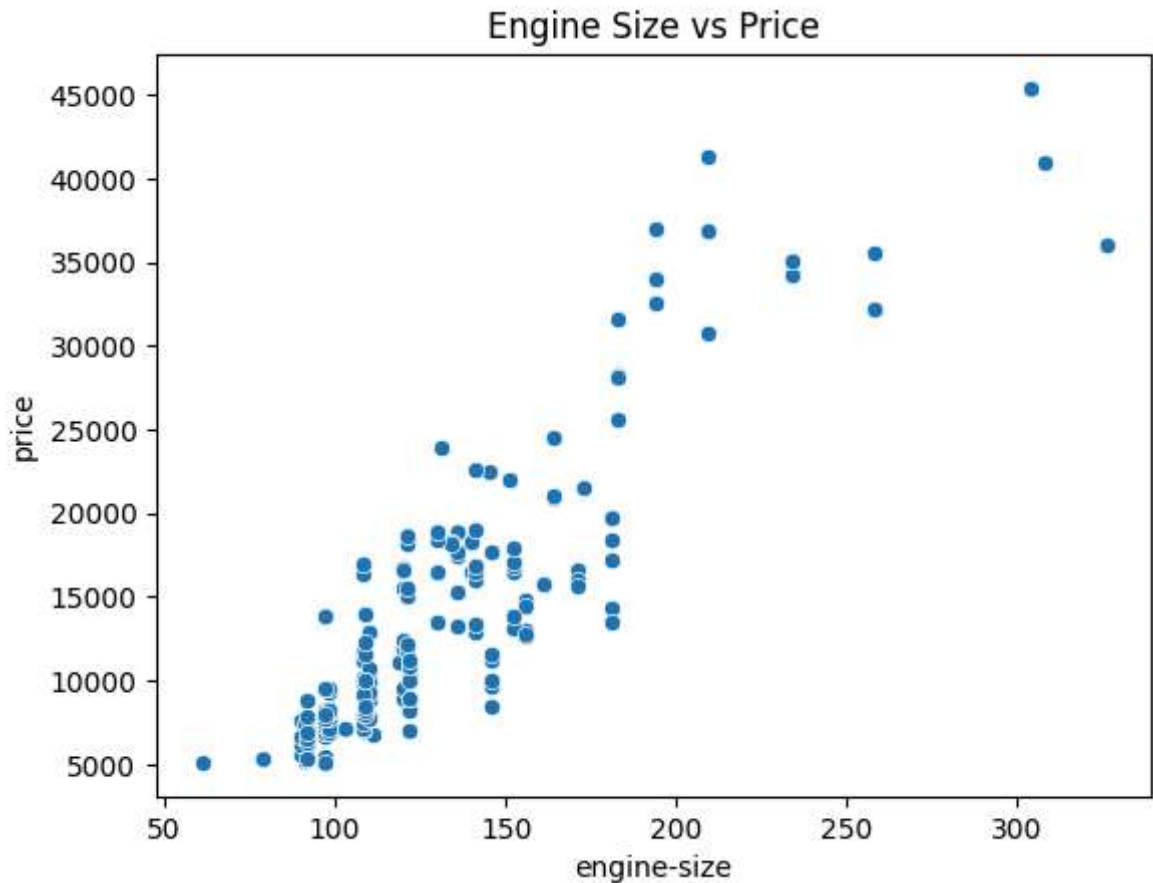
plt.xticks(rotation=45, ha='right')
plt.tight_layout()

plt.show()
```



Insight: Some manufacturers dominate the dataset in terms of number of models, suggesting greater market presence or diversity in product offerings.

```
In [50]: sns.scatterplot(x='engine-size', y='price', data=automobiles_df)
plt.title('Engine Size vs Price')
plt.show()
```



Insight: There is a strong positive relationship between engine size and price, indicating that larger engines significantly increase vehicle cost, suggesting that engine size is a primary driver of pricing across manufacturers.

```
In [51]: cheapest_5 = automobiles_df.sort_values(by='price').head(5)
         cheapest_5
```

Out[51]:

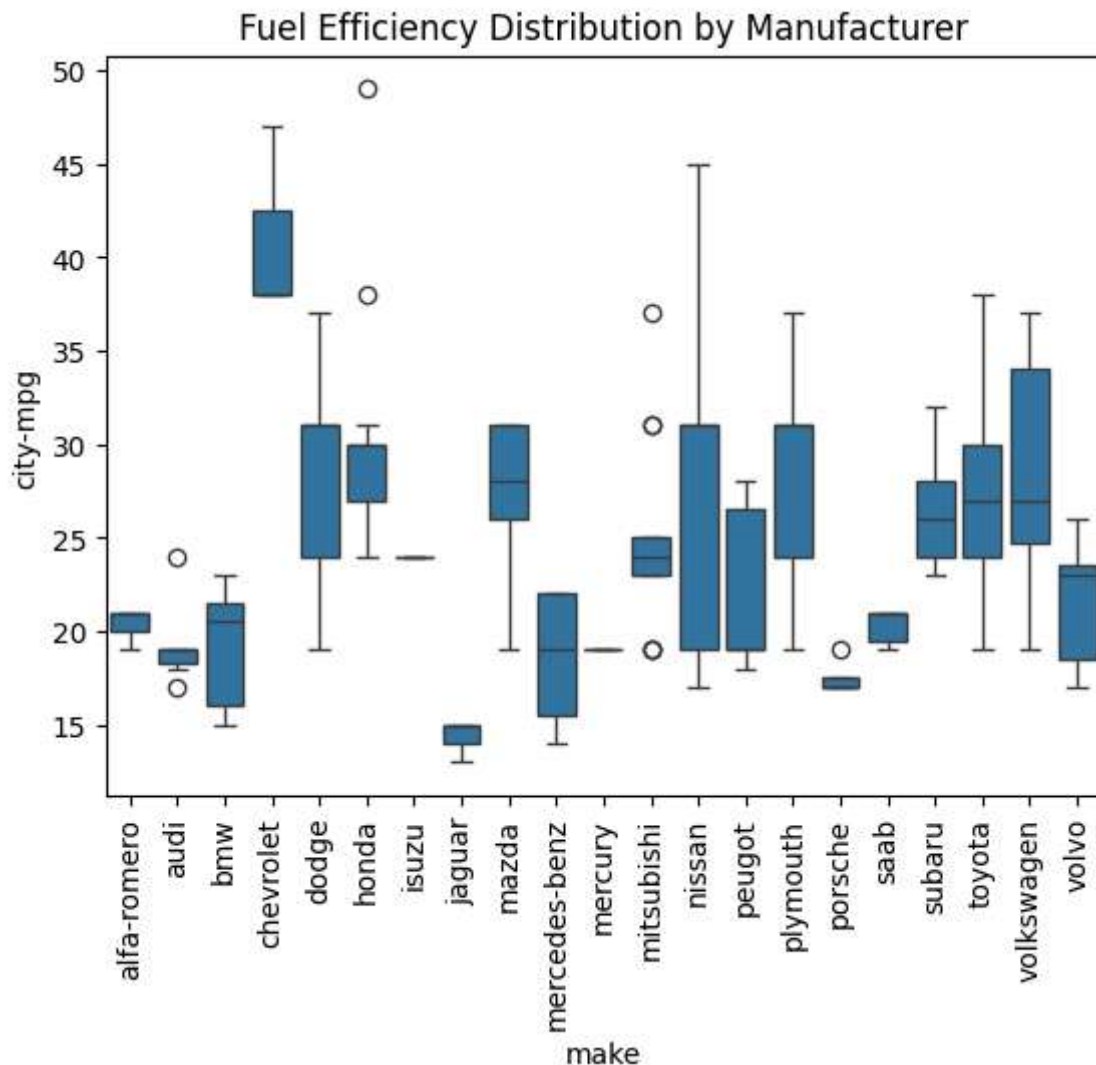
	make	fuel- type	aspiration	num- of- doors	body- style	drive- wheels	engine- location	wheel- base	length	width
138	subaru	gas	std	two	hatchback	fwd	front	93	156	€
18	chevrolet	gas	std	two	hatchback	fwd	front	88	141	€
50	mazda	gas	std	two	hatchback	fwd	front	93	159	€
150	toyota	gas	std	two	hatchback	fwd	front	95	158	€
76	mitsubishi	gas	std	two	hatchback	fwd	front	93	157	€

5 rows × 24 columns



Insight: Cheaper cars generally have smaller engines and better fuel efficiency, highlighting a trade-off between affordability and performance.

```
In [52]: sns.boxplot(x='make', y='city-mpg', data=automobiles_df)
plt.title('Fuel Efficiency Distribution by Manufacturer')
plt.xticks(rotation=90)
plt.show()
```



Insight: Fuel efficiency varies not only between manufacturers but also within them, indicating differences in vehicle types and engine configurations.

Conclusion

The analysis of the automobile dataset reveals several key insights:

- High-priced vehicles tend to prioritise performance over fuel efficiency
- Fuel efficiency varies significantly across manufacturers
- Engine size plays a major role in determining both performance and cost
- Certain manufacturers offer a wider range of models than others

Overall, the dataset demonstrates clear relationships between vehicle features, which can be effectively explored through data cleaning and visual analysis.

